

PRASHANT BAIRAGI

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PROFESSIONAL SUMMARY

Backend-focused engineering student who designed, built, and deployed a production Java/Spring Boot application serving real Android users, hosted on AWS EC2 with Docker. Hands-on across the full backend lifecycle — JWT authentication, REST API design, MySQL data modeling, and cloud deployment. Strong DSA foundation. Seeking to contribute to real engineering work in a product team.

TECHNICAL SKILLS

Languages: Java, SQL, C++, Python

Backend Development: Java, Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, JSON, JWT Authentication, DTO Design, Validation, Exception Handling

Databases: MySQL, PostgreSQL

Cloud & Deployment: AWS EC2, Docker, Vercel

Developer Tools: Git, GitHub, Maven, Postman API Testing, Swagger / OpenAPI, IntelliJ IDEA, VS Code

Core Concepts: OOP, Data Structures, Algorithms, Problem Solving

Additional: Embedded Systems, PCB Design, RF Communication

PROJECTS

Kharcha Pani — *Personal Finance Tracker for Modern India.* [GitHub Repository](#) [Live Documentation](#) **LIVE · IN PRODUCTION**

Production backend powering a live Android finance app with 40+ real-user installs, built with Java, Spring Boot, Spring Security, MySQL, and Docker. [[Android Releases](#)]

- **Production Deployment:** Deployed Dockerized backend to AWS EC2 — configured security groups, networking, and environment secrets for 24/7 availability serving real mobile clients.
- **Security Architecture:** Stateless JWT auth + Spring Security across 12+ REST endpoints, with per-user data isolation enforced at the query layer on every resource.
- **Domain Modeling:** Normalized multi-entity schema (User → FinancialMonth → Expense) via JPA/Hibernate; month-wise budgets with automatic financial-period creation carrying forward prior budget.
- **API Design:** Paginated/sortable endpoints (Spring Data Pageable) hardened with field allowlists against parameter injection; Swagger/OpenAPI documented.

Portfolio Contact API — [GitHub Repository](#) **LIVE**

Backend service powering the contact form of my live portfolio website — Java, Spring Boot, REST APIs, containerised application deployed on EC2, frontend integrated via Vercel.

- Built and Postman-tested validated REST endpoints; submissions delivered instantly as email notifications via Spring Mail (SMTP integration) for zero-maintenance operation.
- Originally persisted to managed PostgreSQL (Neon); migrated to zero-maintenance SMTP delivery via Spring Mail, keeping the complete JPA persistence layer in source for on-demand re-enablement.

Smart Central Locking System — [GitHub Repository](#) **IN DAILY USE · 1+ YR**

Independently designed and deployed an embedded access security system with ATmega328 microcontroller, RF communication (433 MHz), and actuator control. Installed in a personal vehicle and in continuous daily use for over a year.

- **Problem Solving :** Demonstrated and solved real-world access control problem by designing custom solution, firmware, reducing reliance on commercial systems.

EDUCATION

B.Tech in Electronics and Telecommunication Engineering

Jabalpur Engineering College, Jabalpur | 2023–2027

CERTIFICATIONS AND ACHIEVEMENTS

- NPTEL Certifications — Database Management Systems (Elite, 2026), MEMS Fabrication Techniques (Elite, 2025), The Joy of Computing Using Python (2024)
- AWS Cloud Practitioner Essentials (AWS Training & Certification)
- Data Structures & Algorithms — 200+ LeetCode problems solved
- Independent Hardware Deployment — solo-designed embedded security system (ATmega328, RF 433 MHz), running in a real vehicle for 1+ years